Evidence-Guided Compact Multimodal Recognition under Cross-Subject Shift

Team Lazyshu Technical Report for the CUHK-X Small Model Track

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Abstract

We present Lazyshu, a compact solution to the 40-class CUHK-X cross-subject activity-recognition task. A shared R(2+1)D-34 encodes depth, thermal, and infrared video; temporal CNN/recurrent encoders model skeleton and world-oriented IMU signals; and a masked Transformer fuses available modalities. Five full-data fits are merged into one INT8 parameter soup, while compact residual students preserve complementary teacher behavior without a multi-backbone forward ensemble. Separately stored FP16 visual adapters exploit labeled recordings of the same natural person in another environment. Candidate experts are structurally decoded before a deterministic availability router and conservative protection rules select the output. The primary submission achieved 0.92039 public and 0.91176 private accuracy, ranking fourth privately. Its 14 runtime assets total 84,094,248 bytes, and an internal re-execution reproduced both selected CSVs byte-for-byte.

1 Introduction

The CUHK-X Small Model Track evaluates 40-class human activity recognition (HAR) from privacy-preserving visual and sensor modalities under a strict cross-subject split and a 100 MB aggregate runtime-weight limit [7]. Training contains 3,036 labeled clips from users 1–9 and 16–24; the published Testing set contains 405 unlabeled clips from held-out user IDs 10–11 and 25–26. The available modality groups are colorized depth, thermal, infrared (IR), skeleton, inertial measurement unit (IMU), and mmWave radar. Modalities may be missing, corrupted, asynchronous, or recorded by a variable number of devices.

The central difficulty is generalization: several sensor models reached 96–99% training accuracy but only 23–53% on held-out people. Related actions such as reading versus turning pages or sweeping versus mopping depend on subtle temporal and hand–object cues. Every detector, adapter, and decoder state also counts toward the model-size limit.

Our design follows three principles. First, modality encoders should retain appropriate inductive biases: spatiotemporal CNNs for visual motion, temporal CNN/recurrent models for structured sensors, and attention for token interaction. Second, target adaptation should preserve a transferable source function and remain reversible. Third, acquisition structure may propose a correction, but should not override contradictory clip-level evidence.

Our task-specific contribution is the validated composition of shared visual encoding, world-oriented IMU features, explicit availability masks, an INT8 parameter soup, compact residual distillation, protected paired-person adapters, deterministic supervision routing, and evidence-gated structural decoding, together with a complete size, runtime, provenance, and reproduction audit. R(2+1)D, Transformers, MixUp, model soups, FiLM, and adapters are established techniques [6, 8, 9, 11–13]; our contribution is their validated composition for this task, not a claim of inventing those components.

2 Problem Setup and Validation

2.1 Multimodal inputs

Figure 1 summarizes the six input groups without reproducing competition samples. The visual streams differ strongly in appearance, resolution, and failure mode. Skeleton is a short sequence of 17 three-dimensional keypoints. IMU data arrive from up to five body-worn devices with device-specific orientation, while radar contains irregular point detections and timestamps.

We keep raw synchronization and semantic structure separate. Each stream is first resampled on its own temporal support, encoded into tokens, and fused with attention. Acquisition rounds, repeats, and paired-person relations are handled only after clip probabilities are produced. This avoids inventing a one-to-one alignment between heterogeneous sample clocks.

2.2 Dynamic preprocessing

A clip-level union person window is detected with the bundled YOLO11n model [10]; the detector never predicts activity labels. We uniformly sample 32 frames and create aligned 128- and 160-pixel caches. IR contrast is stabilized. Skeleton coordinates are normalized and augmented with bones and temporal velocities. IMU acceleration and angular velocity are rotated into the world frame using device quaternions. Corrupt sampled images fall back to a nearby valid frame; an unusable entire modality is masked out. The original data remain read-only.

2.3 Cross-subject validation

Random clip splitting leaks person and environment identity. We instead use two user-isolated development folds (Fig. 2). Fold 0 holds out users 3, 8, 18, and 23 (652 clips) for architecture selection; Fold 1 holds out users 5, 7, 17, and 22 (705 clips) to test whether gains transfer. Fold 1 began as a blind confirmation fold, but later-stage proposals were checked on it repeatedly. We therefore report both as development validation, not as independent test estimates. Only after a recipe is frozen is its deployment endpoint refitted on all 3,036 labeled clips.

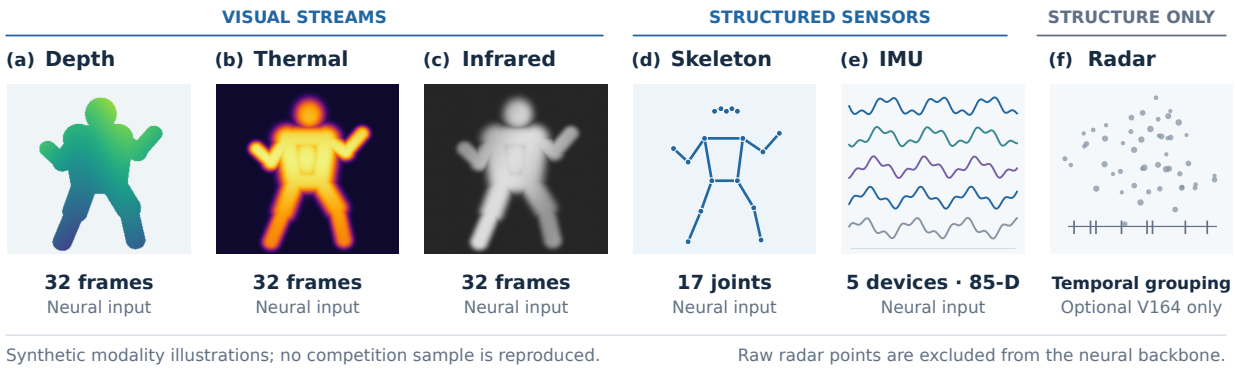


Figure 1: Synthetic schematic of the six modality groups; no competition sample is reproduced. The selected neural backbone consumes visual, skeleton, and IMU signals. Raw radar points are excluded, while radar-only temporal grouping is unique to the optional V164 structural challenger.

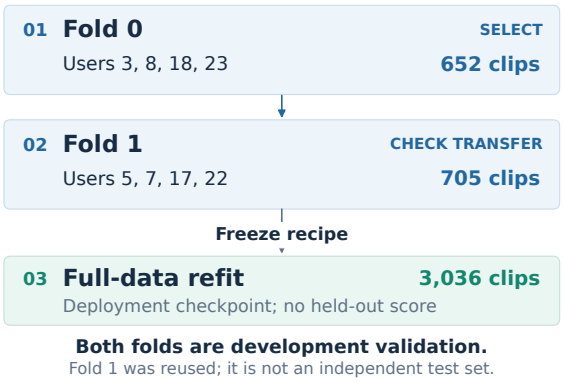


Figure 2: Development protocol. Both folds are development validation; full-data refit begins only after recipe freeze and receives no held-out score.

We distinguish a dataset *user ID* from a natural *person*. Every base validation model excludes the target user and person. In the *same-person-source* stratum, the later adapter may then use labels from that person's released source-environment recording, but never from the held-out target environment; in the *no-same-person-source* stratum, no label from that person is available. Thus paired adaptation measures cross-environment transfer with same-person supervision, not strict unseen-person generalization. A full-data checkpoint is never assigned a held-out score.

3 Method

3.1 Base multimodal network

Figure 3 summarizes the final graph. A shared R(2+1)D-34 network [9], initialized from the public IG-65M-to-Kinetics checkpoint [3] and then fully trained on CUHK-X, encodes three visual streams. The cache has seven channels: three colorized-depth channels, three thermal channels, and one grayscale IR channel. IR is expanded to three channels only inside the shared encoder. Global spatiotemporal maps are projected into 192-dimensional token sequences and also produce stream-level logits.

A temporal CNN encodes normalized skeleton coordinates, bone vectors, and velocities. An 85-feature world-oriented IMU representation is processed by two temporal Conv1D layers followed by a bidirectional LSTM [5]. Learned modality embeddings are added to all sequences. A two-layer pre-normalized Transformer (dimension 192, six heads, feed-forward width 576) processes a class token and all available modality tokens. The padding mask removes missing streams. The fused residual is added to the aggregate visual prediction:

$$\mathbf{z} = \mathbf{z}_{\text{vis}} + h_{\text{fuse}}([\text{CLS}, V_D, V_T, V_{IR}, S, I], \mathbf{m}). \tag{1}$$

The base classifier contains 64,953,671 parameters. Raw radar points are not neural inputs in the selected system.

3.2 Full-data parameter soup

After freezing the architecture, five complementary sources are trained on all labeled clips for fixed 13/16/7/16/9-epoch schedules. They include base, cross-user, progressive-160, scratch masked-distillation, and warm masked-distillation variants. Training uses BF16 autocast, FP32 optimization and exponential moving average, clip-consistent horizontal flips, MixUp [13], label smoothing, and masked-modality distillation [4]. Fold-0-selected soup weights are fixed before full-data fitting:

$$\begin{aligned}\theta_{\text{soup}} = {} & 0.082416\theta_1 + 0.079504\theta_2 + 0.148508\theta_3 \\ & + 0.089572\theta_4 + 0.600000\theta_5.\end{aligned} \tag{2}$$

This is a parameter soup [12], not five forward passes. INT8 storage reduces the base checkpoint to 65.40 MB.

Optimization details. Within the reproduction package, exact base JSONs are in `training/frozen_configs/`; residual and adapter entry points are under `training/source_snapshot/experiments/`. Base fits use AdamW (batch 8, weight decay 10^{-4}, seed 43), linear warmup then cosine decay, and a 0.1 backbone rate factor: cold 128-pixel runs use 3×10^{-4}/two warmup epochs, while warm runs use 1.5×10^{-4}/one. Common settings are MixUp 0.3, smoothing 0.03, EMA 0.99, and masked-distillation weight 0.5

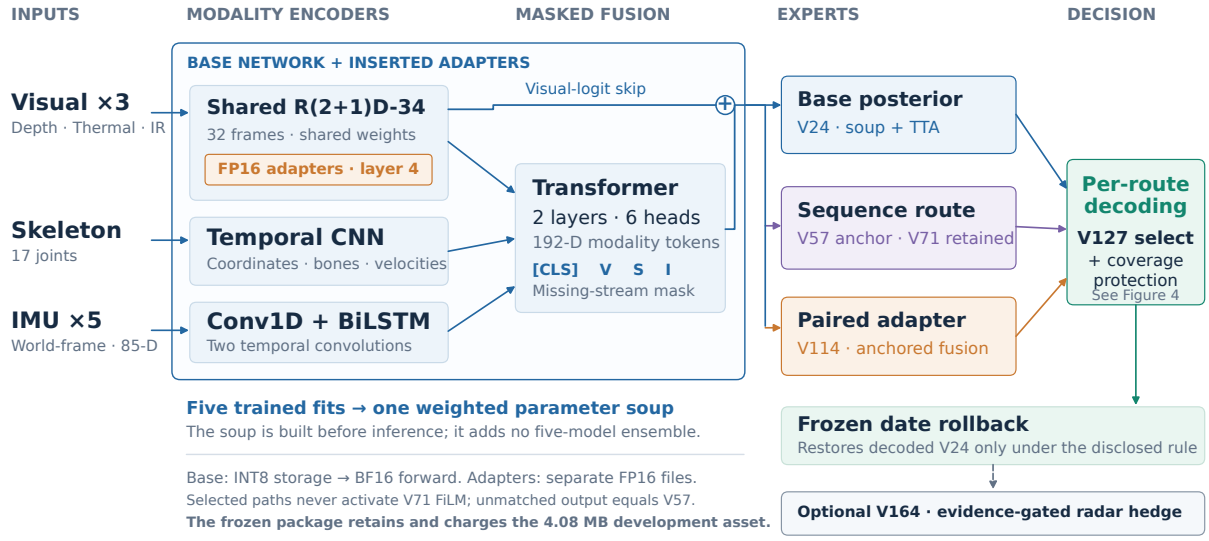


Figure 3: Conceptual inference architecture. The shared encoders and masked fusion network form the INT8-stored base; separately stored FP16 adapters operate inside visual layer 4. Candidate experts are structurally decoded within their routes before V127 selection, coverage protection, the frozen date rollback, and optional V164 evidence gating.

at temperature 2 where enabled. V52/V57 use cross-entropy plus 0.25 retention KL; V71 uses seed 7142, AdamW at 0.003, KL plus 0.2 centered-logit MSE. Final adapters use bottleneck 128, AdamW, batch 16, two epochs, 2×10^{-4}, seed 42, and cross-entropy plus 0.25 retention KL and 0.15 stream distillation at temperature 2.

3.3 Compact residual distillation

Organizer clarification permitted model-generated pseudo-labels on the unlabeled Testing set. No organizer test-label file or manual test annotation was available; the separately disclosed public-leaderboard feedback was not used to train these modules. We therefore adapt only small heads or residual modules while preserving the base representation. The domain-conditioned logit residual (V52) fits zero-initialized 80 → 40 kernel-1 Conv1D corrections to pre/post-adaptation log probabilities. The round-distillation student (V57) groups clips into acquisition rounds and distills a frozen sequence teacher into a 92,392-parameter Transformer [11] that reproduces all 405 teacher decisions. The gated development student (V71) uses a three-layer, 256-dimensional Transformer with a FiLM-style domain gate [8]. Its correction is active only for domain 2—paired source, early environment—and every other domain returns the exact V57 anchor. This internal gate is distinct from the later supervision router. In the two selected submissions, domain-2 rows are instead assigned to the paired adapter, while unmatched rows have the gate off; hence V71's FiLM correction has no direct effect on their labels. The frozen package nevertheless retains, executes, and fully charges this 4.08 MB development asset rather than altering the submitted code path after freeze.

3.4 Protected visual adapters

A rejected modality-specialized backbone replaced the selected model's one shared visual encoder with three branches that shared only lower stages. Unfreezing independent layer 4, or layers 3–4, exposed 117.2 or 178.4 million trainable parameters respectively and caused cross-user forgetting. We therefore freeze the complete source network and insert modality-specific spatiotemporal residual adapters before and optionally after layer 4:

$$x' = x + W_{\text{up}}\,\phi(\text{GN}(\text{DWConv}_{3\times3\times3}(\phi(\text{GN}(W_{\text{down}}x))))). \quad (3)$$

The final projection is zero-initialized, so adaptation starts at the exact source function. GroupNorm avoids target-specific running-statistic drift. Fused multimodal soft targets additionally supervise each visual stream.

CUHK-X includes the same natural people under different user IDs and environments. Validation analogues are users 8 ↔ 18 and 7 ↔ 17; test-time pairs are 10 ↔ 20, 11 ↔ 21, and 26 ↔ 16. The target-only adapter (V98) learns from pseudo-labels and is not promoted. Its paired-person successor (V113) additionally uses true labels from the source-environment recording while retaining refreshed target pseudo-labels. The base checkpoint excludes both IDs of that person before adaptation. Anchored fusion (V114) mixes 60% paired-adapter evidence with 40% of the protected source model (V104). Because no label-source-matched adapter control was run, we attribute later gains to the complete paired-adaptation-and-fusion package, not uniquely to same-person labels.

3.5 Supervision-aware routing and structure

Anchored fusion is selected only with a labeled same-person source; otherwise the supervision router (V127) takes the unmatched V71 code path, whose gate is off and output is exactly

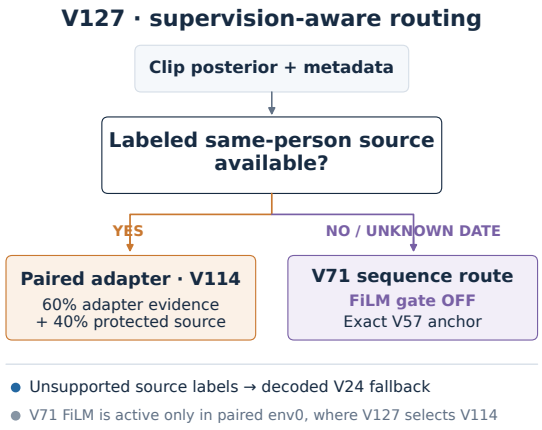


Figure 4: Two-level routing. V127 sends paired rows to V114 and unmatched or unknown-date rows to V71; on the latter route V71's internal FiLM gate is off, yielding the exact V57 anchor.

Table 1: Frozen routing after candidate-specific decoding. "Unsupported" means the decoded V24 incumbent is absent from matched-source label coverage.

Case	Selected output	FiLM	Fallback
Same person, supported	Decoded V114	Not used	–
No same-person source	Decoded V57	Off	V57
Unknown date	Decoded V57	Off	V57
Same person, unsupported	Decoded V24	Not used	V24

V57 (Fig. 4). If the decoded V24 incumbent is absent from matched-source label coverage, protection restores decoded V24. No confidence gate is fitted.

Structural decoder and execution order. For each expert, timestamps first form acquisition rounds. Near-identical rounds of equal length are repeat-pooled; within each round, beam search maximizes clip log-probability plus frozen transition scores while avoiding duplicate labels. When a same-person source exists, a source sequence template may replace the free assignment only if its penalized alignment score is higher. The frozen inference order is:

(1) decode the V24 incumbent, unmatched sequence anchor, and paired adapter separately;
(2) select paired or unmatched decoded output using source availability;
(3) apply matched-source coverage protection, then the disclosed date rollback;
(4) for V164 only, radar timestamps form a separately decoded challenger, accepted by the frozen posterior-rank gate.

Radar-derived temporal bounds can act as a fallback when visual timing is unavailable, but raw radar points are never neural inputs. Radar-only temporal grouping is locally strong but fails online. The evidence-gated radar hedge (V164) therefore keeps the supervision-aware router as the incumbent and accepts a radar-group challenger c only if the original multimodal posterior supports it:

$$\text{accept}(c) \Leftrightarrow \text{rank}_p(c) \le 2 \ \wedge \ \text{rank}_p(y_{127}) \ge 3. \tag{4}$$

This adds eight correct validation clips and no losses across both folds; every held-out user is non-decreasing.

4 Experiments

4.1 Development trajectory

Table 2 and Fig. 5 summarize the principal milestones. Early values are cross-subject classifier ensembles, whereas later values include a frozen structural decoder; the trend should not be interpreted as a perfectly controlled single-model ablation. The largest early gain came from replacing 2D frame encoders with a genuine spatiotemporal video model. Later gains came from compression-aware adaptation, not simply more backbone capacity.

4.2 Adapter and routing ablations

Table 3 isolates the later adaptation path. The target-only adapter improves raw evidence but loses after decoding. The paired adapter with equal fusion (V113) and the retention-heavy anchored fusion (V114) improve the complete package over domain protection, but do not isolate the causal effect of same-person labels. For the selected router, pooled development accuracy is 645/699 (92.27%) with same-person source supervision and 560/658 (85.11%) with none; these are descriptive strata, not independent test estimates.

4.3 Leaderboard results and selection

The selected primary V127 submission (ID 56174717) achieved 0.92039 public and 0.91176 private accuracy, ranking fourth on the Kaggle private leaderboard. The selected secondary V164 submission (ID 56243275) achieved 0.91044 public and 0.90686 private accuracy while changing seven of 405 labels, providing a moderate structural hedge.

The primary CSV includes a conservative rollback chosen through model-level public-leaderboard ablation. The frozen rule first computes the complete protected V127 output, then, only for rows dated 2025-05-31 where V127 disagrees with decoded V24, restores the decoded V24 incumbent. It identifies rows from acquisition date and model disagreement rather than fixed row IDs; on the published 405 clips this yields positions 107, 333, 337, and 350. The rule is explicit in the reproduction entry point, so extra samples on other dates are unchanged and unseen dates follow the unmatched V71/V57 route. This aggregate feedback changed the final primary CSV and is separate from the one-row probes below; no organizer label file or manual annotation was accessed.

One-row public-leaderboard probes. Eleven one-row diagnostics were submitted: three on 11 September before the 12 September primary (base versus sequence teacher), and eight on 13–14 September after recipe freeze (structural challengers). Four dropped exactly 1/201, confirming the incumbent on those public rows; seven caused no score change and therefore did not confirm the proposed candidate label; none improved. These diagnostics only retained incumbents or rejected challengers and changed no training, pseudo-label, transition, threshold, decoder, or radar-hedge rule. By contrast, the earlier aggregate date ablation selected the four-row rollback and did change the primary CSV. These are leaderboard-derived diagnostics, not validation.

Table 2: Major local milestones. V1–V3 report pooled cross-fold accuracy weighted by fold size; later rows show correct clips out of 1,357 and percentage. The selected primary is bold. The full-data deployment checkpoint has no held-out score.

Stage	Main change	Fold 0	Fold 1	Pooled development result
V1	Initial six-modality 2D-CNN/statistical research baseline	54.60%	53.33%	53.94%
V2	Person crops, IR enhancement, world-frame IMU	60.58%	66.10%	63.45%
V3	X3D-S spatiotemporal video expert [2]	66.87%	73.90%	70.52%
V24	Cross-fitted INT8 soup recipe; full-data refit only for deployment	–	–	1,168 / 1,357 (86.07%)
V52	Domain-conditioned logit residual	568 / 652	625 / 705	1,193 / 1,357 (87.91%)
V57	Round-Transformer distillation	567 / 652	628 / 705	1,195 / 1,357 (88.06%)
V71	Gated FiLM development student	570 / 652	631 / 705	1,201 / 1,357 (88.50%)
V127	Supervision-availability routing (selected primary)	569 / 652	636 / 705	**1,205 / 1,357 (88.80%)**
V164	Evidence-gated radar challenger	571 / 652	642 / 705	1,213 / 1,357 (89.39%)
V166	Repeated transition consensus (evaluated online; rejected)	574 / 652	652 / 705	1,226 / 1,357 (90.35%)

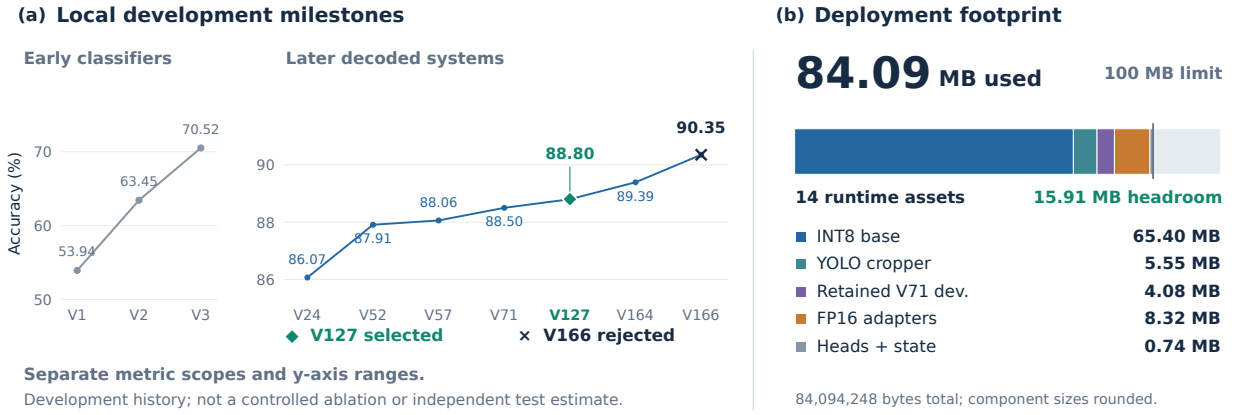


Figure 5: Left: local research progression, with the selected primary in green and the rejected local optimum marked by a cross; early and late metrics have different decoder scope. Right: exact runtime-asset accounting against the decimal 100 MB limit.

Table 3: Later-stage cross-subject ablations. Raw and decoded entries are correct clips out of 1,357. V98 and V113 are distinct adaptation stages.

System	Raw correct	Decoded correct
Fixed multiscale (V89)	1,077	1,170
Protected source adapter (V97)	1,057	1,173
Target adapter (V98)	1,090	1,154
Domain protection (V104)	1,076	1,175
Paired adapter, equal fusion (V113)	1,091	1,176
Anchored adapter fusion (V114)	1,091	1,180
Supervision-aware router (V127)	–	1,205
Evidence-gated radar hedge (V164)	–	1,213

4.4 Negative results and distribution shift

The locally best repeated-transition decoder (V166) reached 1,226/1,357, adding 21 correct clips and no development-fold losses. It remained non-decreasing over 60 partial-script stress views, yet its public score fell to 0.88557 (178/201 versus 185/201 for the primary). Direct radar grouping (V160) similarly reached 1,222/1,357 locally but only 0.84079 publicly. Both were rejected.

Table 4: Selected final submissions. Size counts all 14 runtime assets; time covers all 405 clips on one A800.

System	Public / private	Package / time
Primary router (V127)	0.92039 / 0.91176	84.09 MB / 407.77 s
Radar hedge (V164)	0.91044 / 0.90686	84.09 MB / 407.77 s

Other failures reinforce the same diagnosis. A cross-fitted learned round Transformer dropped the domain-protected anchor (V104) from 1,175 to 1,144; the rejected three-branch visual backbone with 117.2–178.4 million trainable high-level parameters caused immediate held-out-user forgetting; a reproduced official-style IMU CNN-Transformer fitted training near 99% but obtained only 22.85%/24.68% on held-out subjects; and a five-device attention residual regressed on both development folds. The bottleneck is therefore conditional shift across people, environments, objects, and acquisition scripts, not insufficient training capacity.

5 Efficiency and Reproducibility

5.1 Model budget and runtime

The 14 hashed assets total 84,094,248 bytes: a 65.40 MB INT8 base, 5.55 MB cropper, retained 4.08 MB V71 student, 8.32 MB adapters, and 0.74 MB of heads and state. Although V71's FiLM correction is inactive in selected outputs, frozen code still loads and executes it; we retain and charge the full asset. Headroom below the decimal limit is 15,905,752 bytes.

On one A800-SXM4-80GB, 405 clips require 76.21 s preprocessing and 311.17 s neural inference; the remaining 20.39 s covers structural decoding and I/O, for 407.77 s end to end. Peak allocated CUDA memory is 13.49 GB and inference is network-free. The storage target is met; this is not a memory or real-time guarantee.

5.2 Reproduction contract

The entry point discovers the data root, sample count, and modalities; preserves official CSV order; and uses decoded V57 for unseen dates. The four-row rollback follows the frozen date-and-disagreement rule, never loaded row IDs. It verifies every artifact's byte size and SHA-256, then rejects unsupported IDs, schema/order errors, duplicate paths, nulls, or labels outside $[0, 39]$.

INT8 base weights are dequantized for BF16 neural forwards; post-processing uses FP32/FP64 and no stochastic sampling. In an isolated internal audit, both selected IDs were regenerated from raw label-free Testing data on a physical GPU and matched retained CSVs byte-for-byte, including SHA-256. This is not organizer certification.

5.3 External resources and compliance

The public rules forbid manual test annotation and test-ground-truth use [1]. We did not read an organizer test-label file, manually annotate a test sample, or use labels inferred from public-leaderboard feedback to train any model. Archived authenticated organizer replies cover pseudo-labeling, initialization, quantization, and ensemble accounting, but not one-row probes. We therefore retain the disclosures above and describe probes as leaderboard-derived sample information, not organizer-endorsed supervision. No LLM, closed API, or remote service generates training labels or participates in inference. The R(2+1)D conversion repository is MIT-licensed; Ultralytics runtime and YOLO11n are disclosed under AGPL-3.0. Dataset terms and all notices are included.

6 Discussion and Limitations

Three findings generalize. Subject-isolated promotion gates prevented high-capacity regressions; small zero-initialized residuals were safer than broad fine-tuning; and metadata should narrow, not override, multimodal evidence. Limitations remain: transductive modules may transfer imperfectly to new users, unknown dates may differ despite safe fallback, and 128/160-pixel crops lose fine hand–object detail. Repeated TTA and adapter forwards are also heavier than a single edge pass.

7 Conclusion

Lazyshu combines shared spatiotemporal encoding, masked fusion, compact residuals, protected paired-person adapters, and evidence-gated decoding in 84.09 MB. V127 achieved 91.176% private accuracy and fourth place; one offline entry point regenerates its exact output from raw label-free data. Selection jointly considered cross-subject validation, disclosed online evidence, complexity, and reproducibility rather than local score alone.

Acknowledgments

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